

Grade 8
Unit 5
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

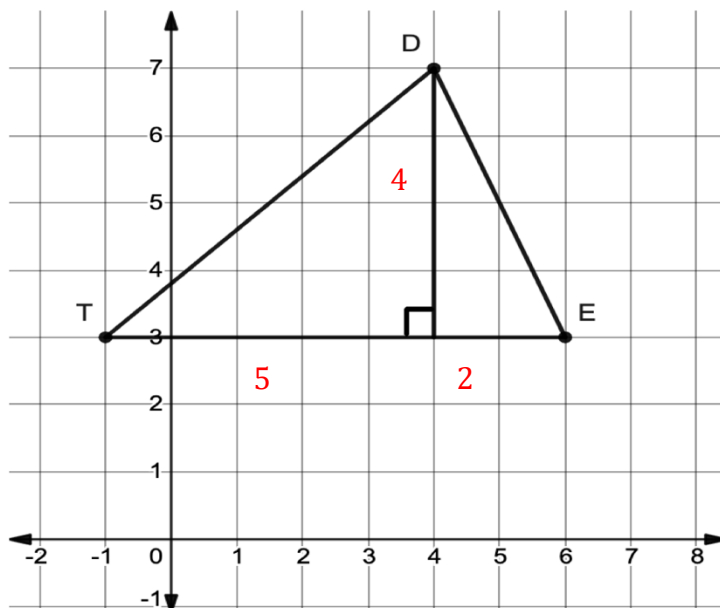
Find the perimeter of Triangle TED .

- How many right triangles make up Triangle TED ?
2 right triangles
- Label all horizontal and vertical lengths on the coordinate plane.
- Find the exact perimeter of Triangle TED .

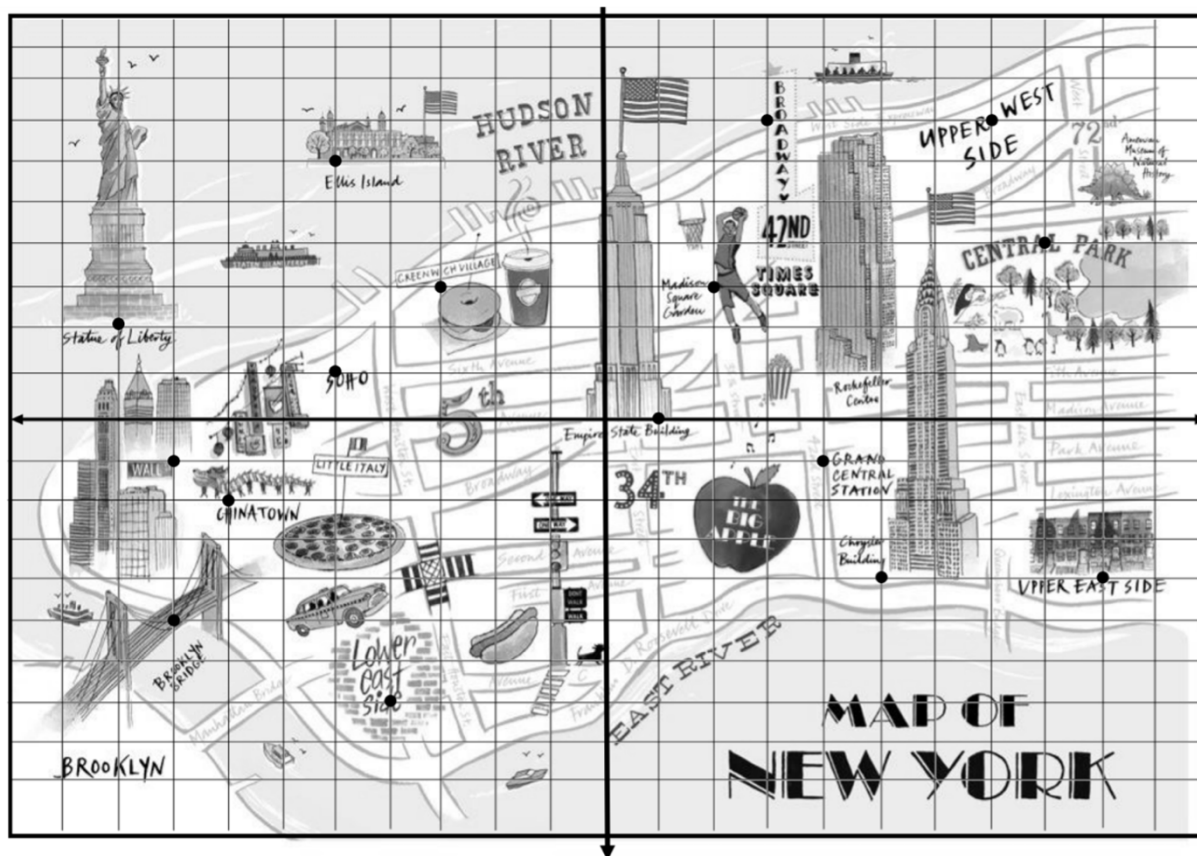
$$\begin{aligned} TD^2 &= 4^2 + 5^2 \\ &= 16 + 25 \\ &= 41 \\ TD &= \sqrt{41} \\ &\approx 6.4 \end{aligned}$$

$$\begin{aligned} ED^2 &= 2^2 + 4^2 \\ &= 4 + 16 \\ &= 20 \\ ED &= \sqrt{20} \approx 4.5 \end{aligned}$$

$$P = 7 + \sqrt{41} + \sqrt{20} \approx 17.9$$



Use the coordinate grid that represents a map of New York City for questions 4–10. Assume 1 unit represents 1 square mile. Round all answers to the nearest tenth.



4. Find the distance between the Upper East Side and Grand Central Station.

$$\begin{aligned}c^2 &= 3^2 + 5^2 & c &= \sqrt{34} \\&= 9 + 25 & &\approx 5.8 \text{ miles} \\&= 34\end{aligned}$$

5. Find the distance between the Statue of Liberty and Broadway.

$$\begin{aligned}c^2 &= 5^2 + 12^2 & c &= \sqrt{169} \\&= 25 + 144 = 169 & &= 13 \text{ miles}\end{aligned}$$

6. Find the distance between Wall Street and Ellis Island.

$$\begin{aligned}c^2 &= 3^2 + 7^2 & c &= \sqrt{58} \\&= 9 + 49 & c &\approx 7.6 \text{ miles} \\&= 58\end{aligned}$$

7. Find the distance between the Chrysler Building and Greenwich Village.

$$\begin{aligned}c^2 &= 8^2 + 7^2 & c &= \sqrt{113} \\&= 64 + 49 & c &\approx 10.6 \text{ miles} \\&= 113\end{aligned}$$

Mia is taking a day trip to New York City. She parks her car near Central Park. She walks to the Empire State building then the Chrysler Building and back to where she parked at Central Park.

8. How far did she walk in one day? Round to the nearest tenth.

$$\begin{aligned}c^2 &= 4^2 + 7^2 & c^2 &= 4^2 + 4^2 & c^2 &= 8^2 + 3^2 & 8.1 + 5.7 + 8.5 &= 22.3 \text{ miles} \\&= 16 + 49 & &= 16 + 16 & &= 64 + 9 \\&= 65 & &= 32 & &= 73 \\c &= \sqrt{65} \approx 8.1 & c &= \sqrt{32} \approx 5.7 & c &= \sqrt{73} \approx 8.5\end{aligned}$$

9. Prove whether the triangle she formed on her walk forms a right triangle.

$$\begin{aligned}65 + 32 &\stackrel{?}{=} 73 & 65 + 73 &\stackrel{?}{=} 32 & 32 + 73 &\stackrel{?}{=} 65 \\97 &> 73 & 138 &> 32 & 105 &> 65\end{aligned}$$

Her walk does form a triangle.

10. If Mia walks at speed of 2.9 miles per hour throughout the day, how will it take her to walk to the Empire State building then the Chrysler Building and back to where she parked at Central Park. Round to the nearest tenth.

$$t = \frac{22.3}{2.9} \approx 7.7 \text{ hours}$$